TOPIC 06

Central Limit Theorem

Why averages become normal, whatever the population looks like, and how to use that fact

WHAT YOU WILL LEARN

- State the Central Limit Theorem (CLT) precisely and in plain words
- See it in simulation for uniform, skewed and bimodal populations
- Know the conditions and the limits (including the n ≥ 30 rule of thumb)
- Use the CLT to compute probabilities for x̄, sums and proportions
- Use the normal approximation to the binomial with a continuity correction
- Avoid the common misconceptions

1. The idea

Take a dataset. It might be normal, uniform, binomial, skewed or something irregular. Draw a sample, compute its mean, and repeat many times. Plot all those sample means. The remarkable fact is that **the histogram of sample means looks like a normal curve**, almost regardless of what the original data looks like. That is the **Central Limit Theorem**, one of the most important results in statistics.

DEFINITION **Central Limit Theorem**

Let $X_1, X_2, \ldots, X_n$ be independent observations from **any** population with mean μ and finite standard deviation σ. For a large enough sample size n, the sampling distribution of the sample mean x̄ is approximately normal:

$$\bar{x} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \qquad \text{equivalently} \qquad Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

Topic 05 already told us the **centre** (μ) and **spread** (σ/√n) of x̄ for *any* n. The CLT adds the **shape**: approximately normal when n is large.

2. A simple demonstration

Following the classroom example, create a dataset of random whole numbers between 1 and 1000 (a roughly **uniform** population, so definitely not a bell shape). Draw samples and look at their means.

Our dataset has 960 numbers, mean 489 and variance 83,372 (theory for a uniform 1 to 1000: mean 500.5, variance about 83,333). We take 30 random samples of n = 25 and compute each mean, then repeat with 1,000 samples to see the shape clearly.

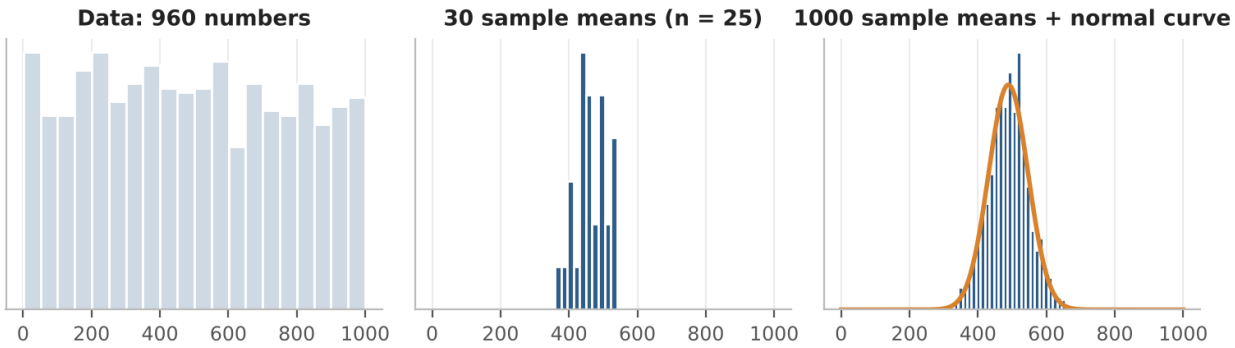


Figure 1. A flat dataset (left) produces bell-shaped sample means (right). The orange curve is the normal distribution predicted by the CLT.

WORKED EXAMPLE **Take 3 samples, compute 3 means**

Sample 1 (first four of its 25 values: 975, 935, 653, 961, ...) has mean 563.0. Sample 2 (starts 602, 586, 537, 603, ...) has mean 556.6. Sample 3 (starts 742, 165, 598, 768, ...) has mean 479.6. Three samples give three different means near 489; repeating this 1,000 times and plotting the means gives the bell curve.

	Value
Mean of the data	488.9
Mean of 1,000 sample means	489.6
SD of 1,000 sample means	58.1
Predicted σ/√n	57.7

3. The CLT in a picture grid

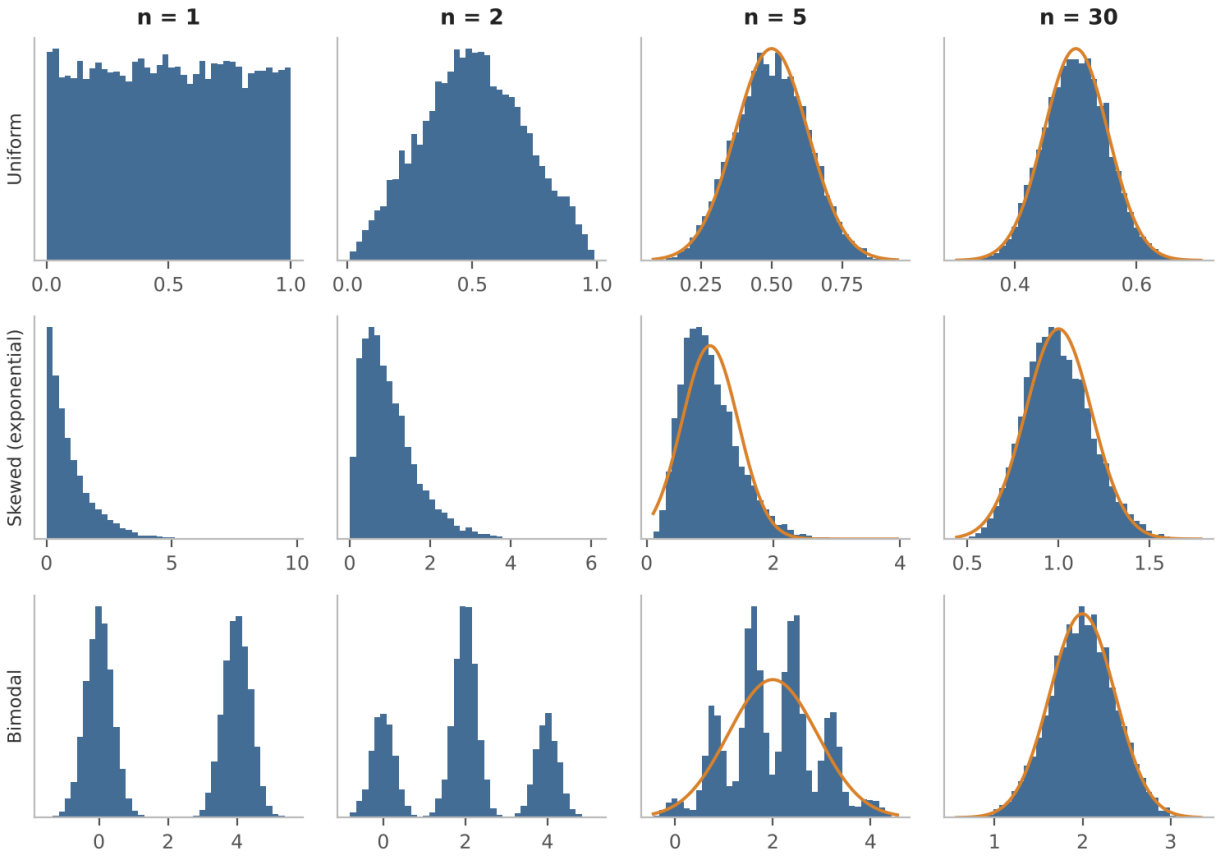


Figure 2. Rows are three very different populations. Columns are the sample size used to form each mean. By n = 30 all rows look normal (orange = CLT prediction); the skewed row needs the larger n most.

Read the figure down the columns. The first column (n = 1) is just the population. As n grows each row smooths into the same bell shape, and the spread shrinks like σ/√n. The skewed population settles slowly. The bimodal population is still lumpy at n = 5 (the two humps are far apart, so the mean of five values lands in one of several clusters) and only looks normal by n = 30. Different shapes need different n, but the destination is always the bell curve.

4. Conditions: when does it work?

Condition	Meaning
Random sample	Observations are drawn at random (or from a randomized experiment).
Independence	One observation does not influence another. If sampling without replacement, n should be below about 10% of the population.
Finite variance	The population must have a finite standard deviation. (Extreme heavy-tailed cases, e.g. the Cauchy distribution, fail.)
Large enough n	The more skewed or heavy-tailed the population, the larger n must be.

How large is 'large enough'?

Population shape	Typical n needed for $\bar{x}$ to look normal
Already normal	Any n (exactly normal)
Roughly symmetric, no outliers	About 10 to 15
Moderately skewed	About 30 (the textbook rule of thumb)
Strongly skewed or with extreme outliers	Often 100 or more

COMMON MISTAKE **n ≥ 30 is a guideline, not a law**

Thirty works well for mildly skewed data. For heavily skewed data (incomes, claim sizes) a sample of 30 may not be enough; look at a histogram of your data first. **Bigger samples give a better normal approximation.**

5. Using the CLT: probabilities for sample means

WORKED EXAMPLE **Package weights**

Packages from a plant have mean μ = 50 g and σ = 12 g, but the distribution is skewed (not normal). An inspector weighs n = 36 packages. What is the probability that their mean exceeds 53 g?

- n = 36 ≥ 30, so by the CLT $\bar{x}$ is approximately normal with mean 50 and SD (standard error) $12/\sqrt{36} = 2$.
- $z = (53 - 50)/2 = 1.50$.
- $P(\bar{x} > 53) = P(Z > 1.50) =$ **0.0668**, about 6.7%.

Without the CLT we could not answer this, because we do not know the exact shape of the population. With it we only need μ, σ and n.

Sums too

The same logic applies to the **total** $T = X_1 + \ldots + X_n$: it is approximately normal with mean nμ and standard deviation $\sigma\sqrt{n}$. A lift certified for 20 passengers whose weights have μ = 75 kg and σ = 15 kg: total mean 1500 kg, SD 67.1 kg, so P(total > 1600 kg) = P(Z > 1.49) = 0.0680.

6. Proportions and the binomial

A count of successes is a **sum** of 0/1 trials and a proportion is their **mean**, so the CLT applies to both. This justifies the normal approximation to the binomial, valid when $np \geq 10$ and $n(1 - p) \geq 10$:

$$X \sim \text{Bin}(n, p) \approx N(np,\ np(1-p))$$

Because a binomial is discrete and the normal is continuous, add a **continuity correction** of ±0.5 so the area covers each integer's whole bar.

WORKED EXAMPLE **Coin flips**

X = heads in n = 100 fair flips, so $X \approx N(50, 5^2)$. Find $P(X \leq 45)$. Exact binomial: **0.1841**. Normal without correction: $P(Z \leq -1) = 0.1587$. With the correction, $P(X \leq 45.5)$: $z = (45.5 - 50)/5 = -0.9$, giving **0.1841**. The corrected answer is much closer to the exact one.

7. Why the CLT matters

- It lets us build **confidence intervals** and **hypothesis tests** for a mean using normal (or t) tables **without** knowing the population distribution.
- It explains why normal distributions appear so often: many real quantities are sums or averages of many small independent effects.
- It tells us **how the standard error behaves**: $\sigma/\sqrt{n}$, the spread of the bell curve of sample means (Topic 07).
- It is the bridge between probability (Topics 02 to 04) and inference (Topics 08 to 11).

8. Common misconceptions

Misconception	Reality
'With a big sample, my data becomes normal.'	No. A large sample's histogram still looks like the population. It is the distribution of the **sample mean** that becomes normal.
'CLT needs a normal population.'	Its power is that it does **not**. Normality of the population is only needed for small samples.
'n = 30 always guarantees normality.'	It is a rule of thumb. Heavily skewed populations need larger n.
'The sample mean equals the population mean once n is large.'	It gets close (spread $\sigma/\sqrt{n}$) but is never exactly equal.
'It applies to any statistic.'	The basic CLT is for sums and means. Medians, maximums and ratios need other results.

9. Verify the CLT in Python

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(0)

n = 30
means = rng.exponential(1.0, size=(10_000, n)).mean(axis=1)   # skewed; mu = sigma = 1

print(means.mean(), means.std())            # ~ 1.0 and ~ 1/sqrt(30) = 0.1826
print(stats.skew(means))                    # small (the population's skew is 2)
print(stats.shapiro(means[:500]).pvalue)    # normality test
```

SUMMARY

- For any population with mean μ and SD σ, $\bar{x}$ is approximately $N(\mu, \sigma^2/n)$ when n is large.
- Standardize with $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$ and use the normal table.
- It works for sums, means and proportions; the binomial approximation uses a ± 0.5 continuity correction.
- The more skewed the population, the larger the n needed; $n \geq 30$ is only a guideline.
- The CLT is about the distribution of $\bar{x}$, **not** the distribution of the data.

Practice questions

Try each question before reading the answer underneath it.

Q1. A population (not normal) has $\mu = 200$ and $\sigma = 40$. Find $P(\bar{x} > 210)$ for $n = 64$.

Answer. SE = 40/8 = 5, z = 2, P = 0.0228.

Q2. Why can we use the normal table in Q1 even though the population is not normal?

Answer. Because n = 64 is large, so by the CLT $\bar{x}$ is approximately normal.

Q3. A fair coin is tossed 200 times. Approximate $P(X \geq 110)$ with the continuity correction.

Answer. $X \approx N(100, 7.07^2)$. $P(X \geq 109.5)$: z = 1.344; probability 0.0896. (Exact: 0.0895).

Q4. State the three facts the CLT and Topic 05 give about the sampling distribution of $\bar{x}$.

Answer. Centre μ, spread $\sigma/\sqrt{n}$, and shape approximately normal for large n.

Q5. n = 12 values come from a population that is extremely right-skewed. Is the CLT safe to use?

Answer. Probably not; n is small and the population is strongly skewed. Use a larger sample or a method that does not assume normality.

Q6. Individual daily sales have $\mu = 500$, $\sigma = 100$ (not normal). Over 25 days, what is the probability that the average exceeds 520?

Answer. SE = 100/5 = 20, z = 1, P = 0.1587.